Lattice-Based Rate-Limited Tokens in the Random Oracle Model

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Motivation

Rate-Limited Access



Bob

Article ; cookie / account ID = Bob

Article, if count[Bob] $< N$



News Outlet

reader	reads
Bob	$3/N$
Alice	N/N
Carol	$1/N$

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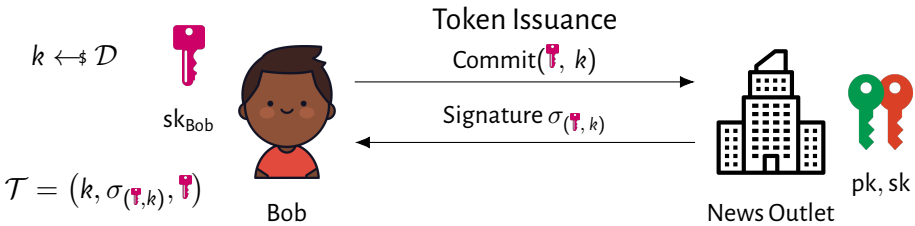
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News Outlet knows *who* the user is.

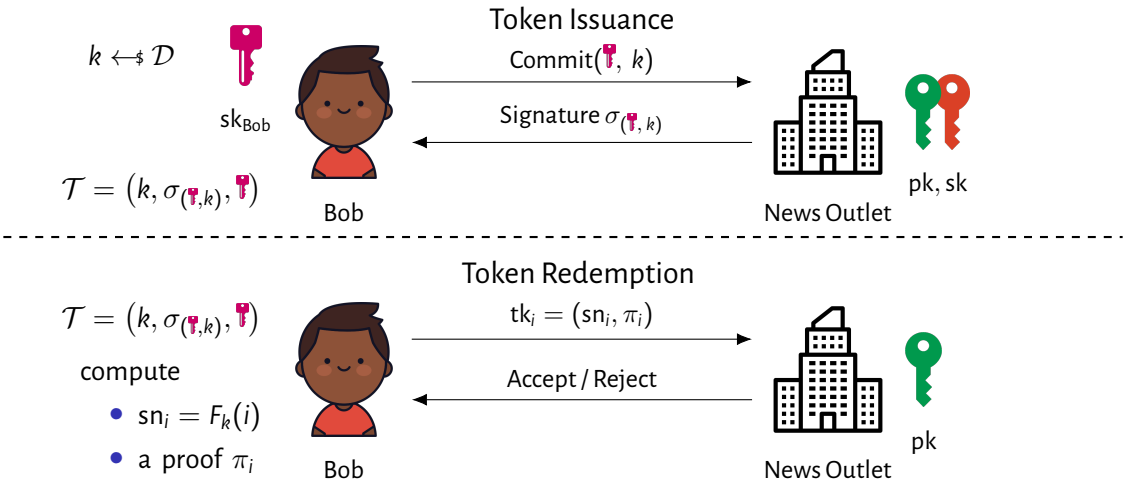
Anonymous Rate-Limited Tokens

[CHL05; CHK+06; CDL+25]



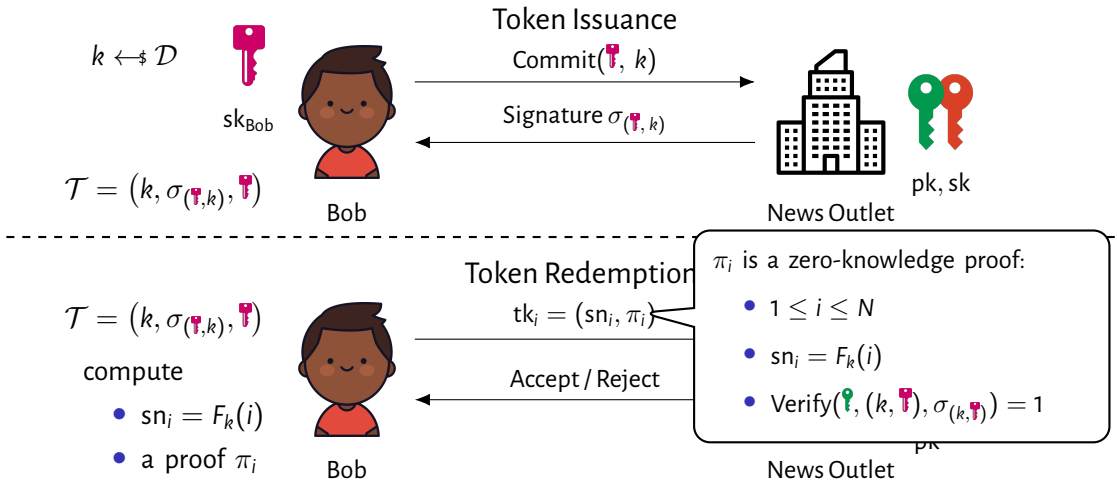
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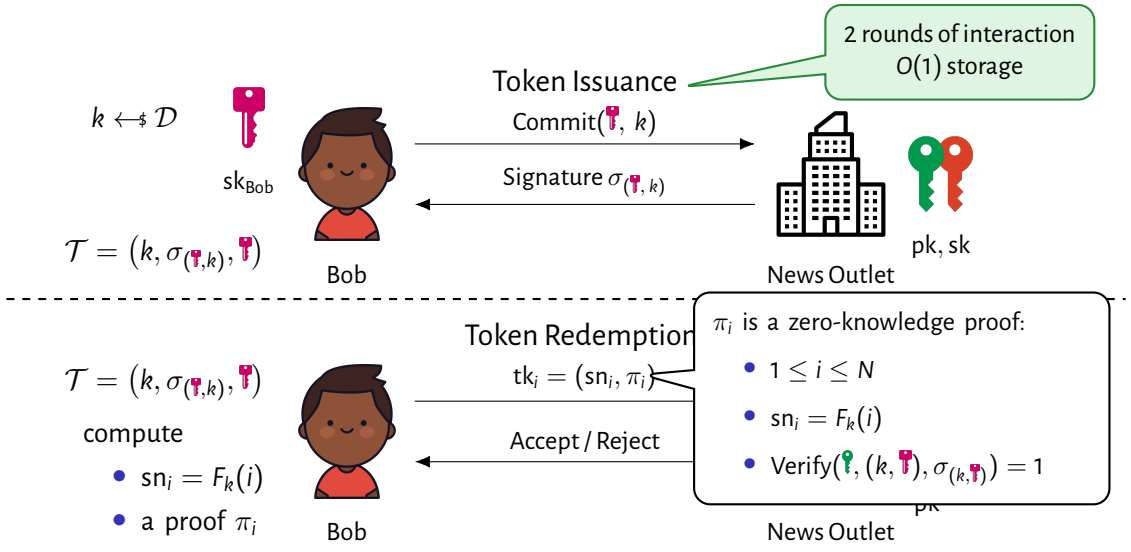
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Anonymous Rate-Limited Tokens

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- **Anonymity**: Tokens are unlinkable to each other and to issuance.
- **Soundness/ One-more Unforgeability**: A user cannot redeem more than N distinct tokens.
- **Exculpability**: Honest users can never be framed for double-spending.
- **Linkability**: We can always recover the identity of a double spender.

In practice:

- Apple: Anonymous Rate-Limited Credentials (IETF draft)
- Cloudflare: rate-limiting bots and AI agents
- Google: Anonymous Counting Tokens

Double-spending occurs if we have $\mathsf{tk}_i = (\mathsf{sn}_i, \pi_i)$, $\mathsf{tk}_j = (\mathsf{sn}_j, \pi_j)$ such that $\mathsf{sn}_i = \mathsf{sn}_j$

Anonymous Rate-Limited Tokens

Prior Works vs. Ours

Scheme	**Token Size** (Estimate)	**Anonymity**	**Assumptions**
[CHL05]	$<$ 50KB	Computational	ROM + Strong DDH Inversion
[CDL+25]	$<$ 50 KB	Unconditional	GGM + q-ARSDH + SXDH
[BCK+09]	$<$ 50 KB	Computational	DLIN + Strong DDH Inversion + Triple DH
[LN26]	$<$ 50 KB	Computational	GGM + Strong DH Inversion
[LLN+17; DLN+20]	$>$ 3 MB* Unless you use a SNARG	Computational	ROM + LWE + SIS

- ROM: Random Oracle Model
- GGM: Generic Group Model
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Ours	$<$ 1 MB* For N=256	Computational	ROM + LWE + SIS

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Focus:

- Linkability
- Soundness

- ROM: Random Oracle Model
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Our Construction: Lattice-Based Rate-Limited Tokens

Pseudo-Random Function Issues

Standard model [BP14; BLM+13; Kim20]

$$F_{\mathbf{a},\mathbf{B}}(x) = \left\lfloor \mathbf{a}^\top g(\mathbf{B}, x) \right\rceil_p$$

- q is superpolynomial in magnitude.
- Proving a correct evaluation of F implies proving g, a large arithmetic circuit.

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Random oracle model [BLM+13]

$$F_{\mathbf{k}}(x) = \left\lfloor \mathbf{k}^\top H(x) \right\rceil_p$$

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- No g: H returns random matrices.
- Security from Learning with Rounding:

$$\left(\mathbf{A}, \left\lfloor \mathbf{k}^\top \mathbf{A} \right\rceil_p\right) \overset{c}{\approx} (\mathbf{A}, \mathcal{U}(\mathbb{Z}_p^m))$$

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Our inputs are just counters $i \in \{1, \ldots, N\}$, so a **weak PRF** suffices.

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We can always recover the identity of a double spender.

Double-spending occurs if we have $\mathsf{tk}_i = (\mathsf{sn}_i, \pi_i)$, $\mathsf{tk}_j = (\mathsf{sn}_j, \pi_j)$ such that $\mathsf{sn}_i = \mathsf{sn}_j$, where the sns are outputs of pseudo-random function. However, if tk_i and tk_j belong to different users and $\mathsf{sn}_i = \mathsf{sn}_j$ with high probability, this is a problem.

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Given $F_{\mathbf{k}}(x) = \left\lfloor \mathbf{k}^\top H(x) \right\rfloor_p$, we cannot rely on a user to honestly sample $\mathbf{k}$, e.g., taking $\mathbf{k} = \mathbf{0}^n$ guarantees a collision with any token $\mathsf{tk}^* = (\mathsf{sn}^*, \pi^*)$, where $\mathsf{sn}^* = 0$.

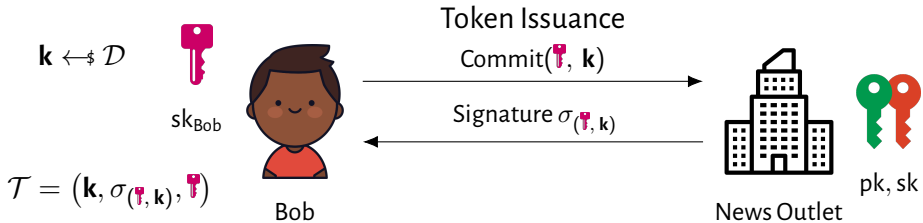
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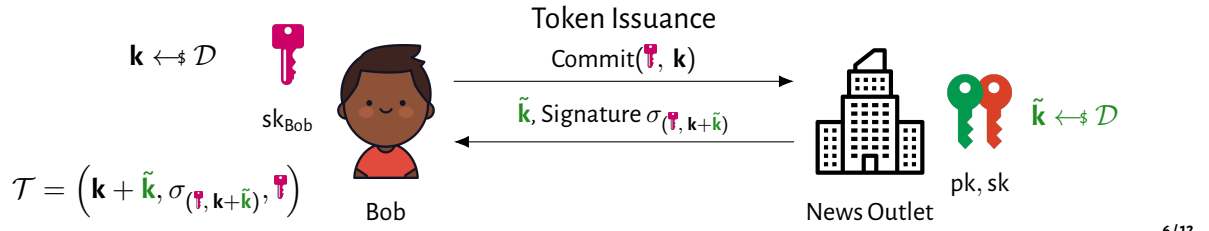
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Prior work [LLN+17; DLN+20] relied on N being polynomial and $\mathbf{k} + \tilde{\mathbf{k}} \sim \mathcal{U}(\mathbb{Z}_q^n)$ to argue that $F_{\mathbf{k}+\tilde{\mathbf{k}}}$ is *collision resistant*. But uniform keys lead to large proofs.

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Our Solution: $F_{k_1+k_2}(x) = F_{k_1}(x) + F_{k_2}(x)$

Lemma (Informal)

Let F be a secure key-homomorphic PRF with a polynomial-size domain $\mathcal{X}$ and a super-polynomial range $\mathcal{Y}$. If every key has the form $k_1 + k_2$, where k_2 is sampled honestly, then for keys of different users and all $x_1, x_2 \in \mathcal{X}$, $F_{k_1+k_2}(x_1) \neq F_{k_1'+k_2'}(x_2)$ with overwhelming probability.

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- No restriction on keys' distributions: any high min-entropy $\mathcal{D}$ works as long as F is secure. Taking $\|\mathbf{k}\|_\infty < q/2$ gives smaller proofs.
- Key-homomorphism can be relaxed to concatenation, i.e., $F_{k_1}(x) \| F_{k_2}(x)$.

Our Construction: Lattice-Based Rate-Limited Tokens

Soundness

The number of tokens N is public. Therefore, we can carve out the random oracle from the proof π by rewriting the statement as:

$$\exists i \in \{1, \ldots, N\}, \mathsf{sn}_i = F_k(H(i))$$

where $H(i)$ is provided by the verifier (*NewsOutlet*).

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More concretely, setting $\mathbf{X}_i = H(i)$, where $\mathbf{X}_i \in \mathbb{Z}_q^{n \times m}$, we want to prove:

$$\text{Given } \{\mathbf{X}_1, \ldots, \mathbf{X}_N\} \text{ and } \mathbf{y} \in \mathbb{Z}_q^m, \text{there exist } \mathbf{k} \in \mathbb{Z}_q^n \text{ with } \|\mathbf{k}\| \leq \beta$$
$$\text{and } i \in \{1, \ldots, N\} \text{ such that } \mathbf{y}^\top = \mathbf{k}^\top \mathbf{X}_i$$

This is a Disjunctive/OR-statement that we know how to prove using a Σ-protocol [Lyu12].

Σ-Protocol

Disjunctive Statement [GGH+22]

To prove knowledge of $\mathbb{w}$ such that $R_{\mathcal{L}}(\mathbb{x}_i, \mathbb{w}) = 1$ for *one* of $\mathbb{x}_1, \ldots, \mathbb{x}_N$, Goel et al. [GGH+22] use a 1-*out-of-*ℓ *partial-binding vector commitment*:

$$v = \boxed{a_1 \mid a_2 \mid \cdots \mid a_\ell} \xrightarrow{\text{commit}} \text{cmt}, o$$

$$v' = \boxed{a'_1 \mid a_2 \mid \cdots \mid a'_\ell} \xrightarrow{\text{equivocate}(\text{cmt}, o)} o'$$

$$\text{open}(\text{cmt}, v, o) = 1 \wedge \text{open}(\text{cmt}, v', o') = 1$$

$$o \approx o'$$

Security: Can only equivocate on $\ell - 1$ positions.

Σ-Protocol

Disjunctive Statement [GGH+22]

$\mathcal{P}$ $\qquad$ $\mathbb{x}_1, \mathbb{x}_2, \ldots, \mathbb{x}_N \in \mathcal{L}$, $R_{\mathcal{L}}(\cdot, \cdot)$ $\qquad$ $\mathcal{V}$

$\mathbb{w}$
s.t. $\exists i \in \{1, \ldots, N\}$
$R_{\mathcal{L}}(\mathbb{x}_i, \mathbb{w}) = 1$



- **Compute** a_i
- **Sample** $a_j \xleftarrow{\$} \{0,1\}^t$**, for** $j \neq i$
- cmt $\leftarrow$ commit$(a_1, \ldots, a_i, \ldots, a_N)$

cmt →
← c



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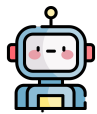


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cmt $\longrightarrow$
c $\longleftarrow$

Compute z_i



$a'_j \longleftarrow$ (c, z_i)

- **Compute** $o' \leftarrow$ equivocate$\big(\text{cmt}, (a'_1, \ldots, a_i, \ldots, a'_N), o\big)$

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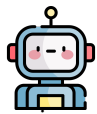


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o', z_i $\rightarrow$

Proof size is $O(\log N)$

Goel et al. partial-binding vector commitment is based on the discrete logarithm assumption, which is not post-quantum secure

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Partial-Binding Vector Commitment from Lattice Assumptions

Case of $1 - 2$ **binding.**

Public Parameters: $\mathbf{A}, \mathbf{B} \in \mathbb{Z}_q^{n \times m}$.

- $\mathrm{Gen}(b \in \{0,1\})$: Sample a low-norm matrix $\mathbf{T}$;
  Then compute $\mathbf{D}_{1-b} = \mathbf{A} \cdot \mathbf{T}$; $\mathbf{D}_b = \mathbf{B} - \mathbf{D}_{1-b}$.

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  $$\mathbf{t} = \mathbf{A}\mathbf{r} + \mathbf{D}_0\mathbf{v}_0 + \mathbf{D}_1\mathbf{v}_1$$

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  $$\mathbf{r}' = \mathbf{r} + \mathbf{T} \cdot \boldsymbol{\Delta}, \qquad \boldsymbol{\Delta} = \mathbf{v}_{1-b} - \mathbf{v}'_{1-b}$$

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Correctness of Equiv

$$\mathbf{A}\mathbf{r}' + \mathbf{D}_b\mathbf{v}_b + \mathbf{D}_{1-b}\mathbf{v}'_{1-b} = \mathbf{A}\mathbf{r} + \mathbf{D}_{1-b}(\mathbf{v}_{1-b} - \mathbf{v}'_{1-b}) + \mathbf{D}_b\mathbf{v}_b + \mathbf{D}_{1-b}\mathbf{v}'_{1-b} = \mathbf{A}\mathbf{r} + \mathbf{D}_b\mathbf{v}_b + \mathbf{D}_{1-b}\mathbf{v}_{1-b} = \mathbf{t}$$

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Partial-Binding Vector Commitment from Lattice Assumptions : Security

Case of $1-2$ binding.

Let $\mathcal{A}$ be a PPT adversary that given $\mathbf{A}, \mathbf{B}$ outputs $(\mathbf{D}_0, \mathbf{D}_1, \text{cmt} = \mathbf{t})$ and $(\mathbf{v}_0, \mathbf{v}_1, \mathbf{r})$ and $(\mathbf{v}'_0, \mathbf{v}'_1, \mathbf{r}')$ such that $\mathbf{v}_0 \neq \mathbf{v}'_0$ and $\mathbf{v}_1 \neq \mathbf{v}'_1$.

$$\mathbf{t} = \mathbf{A}\mathbf{r} + \mathbf{D}_0\mathbf{v}_0 + \mathbf{D}_1\mathbf{v}_1 = \mathbf{A}\mathbf{r}' + \mathbf{D}_0\mathbf{v}'_0 + \mathbf{D}_1\mathbf{v}'_1$$

$$\implies [\mathbf{A} \mid \mathbf{D}_0 \mid \mathbf{D}_1] \begin{bmatrix} \mathbf{r} - \mathbf{r}' \\ \mathbf{v}_0 - \mathbf{v}'_0 \\ \mathbf{v}_1 - \mathbf{v}'_1 \end{bmatrix} = 0$$

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However, this is not complete: we do not know whether $\mathbf{D}_0 = \mathbf{A}\mathbf{T}$ or $\mathbf{D}_1 = \mathbf{A}\mathbf{T}$. To solve this, $\mathcal{A}$ also provides a proof of

$$\mathbf{D}_0 = \mathbf{A}\mathbf{T} \ \vee \ \mathbf{D}_1 = \mathbf{A}\mathbf{T}$$

Summary of Our Contributions

1. Lattice-based rate-limited tokens in the random oracle model.
2. Lattice-based partial-binding vector commitment.
3. A batch proof for conjunctive normal form of

$$\bigwedge_i \left(\mathbf{s}^\top \mathbf{A}_{i,0} = \mathbf{t}_i^\top \vee \mathbf{s}^\top \mathbf{A}_{i,1} = \mathbf{t}_i^\top\right)$$

4. Lattice-based Anonymous Counting Tokens.
5. New approaches to handle **linkability** and **exculpability** in the lattice setting

https://ia.cr/2026/1649



Some icons were sourced from Flaticon.com

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